

Quantifying HPC Utilization for Optimized Resource Deployment

Introduction

Most studies of HPC usage have the end goal of justifying institutional investment in HPC resources to administrators and policy makers. Various studies have demonstrated that HPC investment does indeed contribute to institutional output at different levels. In their study measuring the productivity of computational researchers vis a vis non computational researchers, Smith, 2022¹ showed that HPC investment leads to statistically significant increases in research output for the computational researchers when compared to the non computational researchers. This holds true on average across different research fields and is not confined solely to the traditionally computationally intensive research areas. A number of studies have also been carried out analyzing usage statistics with different tools, including XDMOD² to paint a picture of the kinds of jobs researchers run on HPC systems. These statistics include cpu usage, memory usage, gpu consumption, processing times, wait-times and other utility metrics.³

All these metrics are important in demonstrating usage patterns and justifying the availability of the resources to the institution. As more and more researchers recognize the importance of integrating computational resources in their research, it also happens that more researchers without the requisite background are using these resources. They are immediately faced with a number of challenges including how to properly adapt codes from desktop environments, familiarizing themselves with HPC environments, correctly setting HPC job parameters and just generally using HPC resources productively.

By studying the usage behavior of the computational researchers through the HPC usage metrics and analyzing the most common issues faced by researchers as they port their work and run on HPC systems we can determine and anticipate user needs. Knowing this, we can plan how best to deploy limited resources (personnel) for optimized compute resource usage.

Classifying Research Computing Resources

Generally speaking, the resources at a HPC center can be classified into two: Compute Resources and Personnel that manages and facilitates the compute resources for researchers. The personnel can be further divided into different sub-categories based on the CaRCC research computing and data facings^{4,5}:

¹ Preston Smith. 2022. Measuring the Relative Outputs of Computational Researchers in Higher Education. In Practice and Experience in Advanced Research Computing (PEARC '22). Association for Computing Machinery, New York, NY, USA, Article 21, 1–7. <https://doi.org/10.1145/3491418.3530764>

² Thomas R. Furlani, Barry L. Schneider, Matthew D. Jones, John Towns, David L. Hart, Steven M. Gallo, Robert L. DeLeon, Charng-Da Lu, Amin Ghadersohi, Ryan J. Gentner, Abani K. Patra, Gregor von Laszewski, Fugang Wang, Jeffrey T. Palmer, and Nikolay Simakov. 2013. Using XDMoD to facilitate XSEDE operations, planning and analysis. In Proceedings of the Conference on Extreme Science and Engineering Discovery Environment: Gateway to Discovery (XSEDE '13). Association for Computing Machinery, New York, NY, USA, Article 46, 1–8. <https://doi.org/10.1145/2484762.2484763>

³ Hammond, John. "TACC_stats: I/O performance monitoring for the intransigent." Invited Keynote for the 3rd IASDS Workshop. Vol. 2011. 2011.

⁴ 2018. Research Computing and Data Professionals Job Elements and Career Guide. <https://carcc.org/wp-content/uploads/2019/01/CI-ProfessionalizationJob-Families-and-Career-Guide.pdf>

⁵ Daphne McCanse. 2022. Understanding the CaRCC Facings. Online. <https://carcc.org/2022/10/03/understanding-the-carcc-facings/>

- Researcher Facing
- System Facing
- Software/Data Facing
- Sponsor/Stakeholder Facing

The compute resources are quite simply the supercomputing cluster cpus, gpus and storage both onsite and cloud that are available to the researchers depending on their institutions.

This study focuses on the researchers at George Mason University and the compute resources as managed by the George Mason University Office of Research Computing. Below are the details of the computing resources (compute and personnel) currently available at Mason:

Clusters⁶:

- Hopper: *"11840 core – primary compute cluster, which provides support for scheduled workloads via Slurm. The cluster has two high-speed networking fabrics: a redundant Ethernet network comprising 100 Gbps spine switches and 25 Gbps leaf switches, and an HDR Infiniband network providing 100 Gbps to each node. Users access HOPPER via ssh connections to login nodes or the new ORC Open OnDemand server, which provides access via a web interface to interactive applications such as Matlab and JupyterLab. A flash based VAST storage cluster provides fast scratch storage on HOPPER."*
- Argo - *" 2000 core Linux cluster also providing a scheduled job environment using Slurm. There are a number of different node configurations on ARGO ranging from 16 cores to 32 cores per node. Each node provides a minimum of 4 GB/core, however it has some nodes with 512 GB of RAM and one node with 1.5 TB of RAM. Additionally there are 4 nodes each with 4x NVIDIA 32GB V100, two nodes support NVLink. There are also four nodes with NVIDIA K80 GPUs totaling 20 K80 GPU devices. ARGO supports three workload domains – General, Large Memory, and GPU. The network interconnect is provided by an FDR Infiniband fabric (56 Gbps)."*

Personnel:

- Researcher Facing [3 computational scientists and 2 GRAs]
- System Facing [3 system Engineers]
- ~~Software/Data Facing~~
- Sponsor/Stakeholder Facing [1 Jayshree]

Tracking Resource Usage (Researcher Facing)

To determine researcher behavior on the Mason clusters, we use data extracted from SLURM logs between January 2020 and June 2022 and generated graphs to visualize different details including

⁶ <https://orc.gmu.edu/resources/computing-systems/computing-resources/>

Cluster Statistics

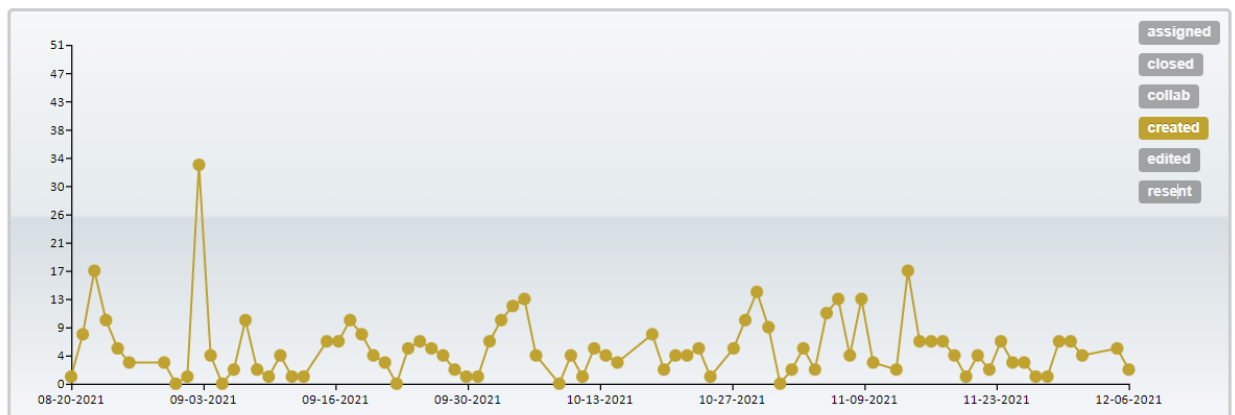
These give a clearer picture on how the cluster is actually being used. From the extracted data, we can look at the number of jobs being run over different periods, types of jobs:cpu intensive vs you intensive jobs, processing times, memory requests, exit states, departments of origin and which groups are the heaviest users.

Tickets

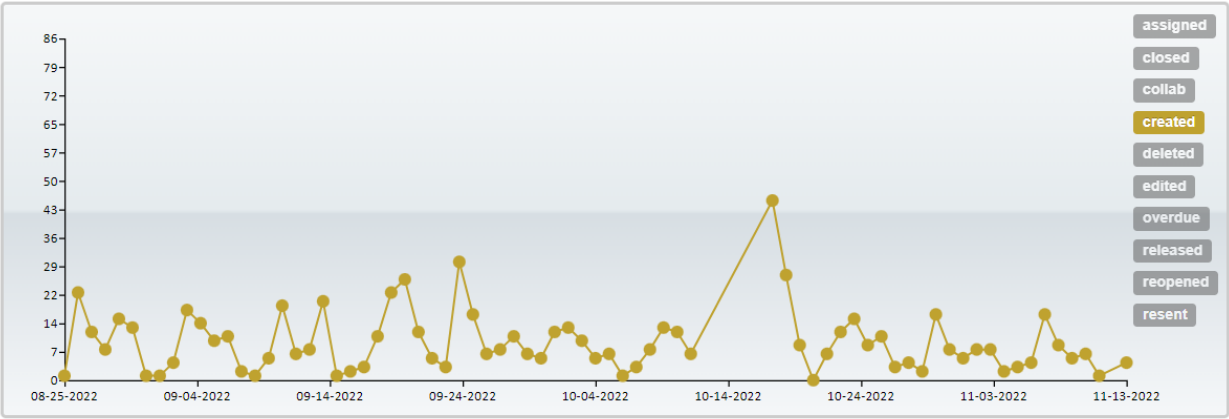
We can also investigate the tickets and questions being asked to get a better understanding of the researchers' challenges. From the ticketing system, we can determine cycles in the frequency of issues being raised, and by analyzing the text of the tickets themselves, figure out knowledge gaps, common asks and requests [and responses].

The two figures below show tickets opened in a similar time period for Fall semesters in 2021 and 2022. For the 2022 case, the spike around 10/14 is attributable to the os-ticket email outage. Ultimately, it should be possible to map any cycles in the tickets over different periods including the summer and spring terms.

Ticket Activity ?



Ticket Activity ?



For better insights, it would be necessary to have better tags to the tickets so we can easily classify the issues being raised. An obvious start would probably be correctly tagging the account creation tickets so they can easily be separated from the rest of the tickets. This would also make it easier to track new account creation requests. The current tags are shown in the image below:

Statistics ?

Statistics of tickets organized by department, help topic, and agent.

Range: October 14, 2022 - November 14, 2022 (America/New_York)

	Department	Topics	Agent						
Help Topic		Opened	Assigned	Overdue	Closed	Reopened	Deleted	Service Time	Response Time
		?	?	?	?	?	?	?	?
	General Inquiry	4	0	0	2	0	0	31.0	11.8
	Report a Problem	6	1	0	3	0	1	719.2	16.6
	orchelp_at_gmu.edu	207	52	0	115	1	3	158.0	22.3
	New ARGO Account	1	0	0	0	0	0	0.0	0.0
	Internal Ops	1	1	0	0	0	0	0.0	23.0

A text analysis of the ticket content would also be useful. This can be achieved by extracting the text from the ticket database and running various analyses on them.



Matching Needs and Resources to project resource needs

Ultimately, by having a better understanding of the usage behavior of researchers and having a clearer view of their needs, we can do two things:

- Have a ready arsenal for tackling the most commonly faced issues.
- Be more strategic in how the Computational Specialists spend their time.

This will hopefully lead to better utilization of the cluster compute resources by researchers.

Final thoughts

1. A great obstacle for new users is working on the cluster from the shell or in terminal sessions, writing SLURM scripts and submitting jobs in batch mode. As users are increasingly being made aware of the availability of the Open OnDemand GUI for accessing the cluster, it will likely affect usage trends positively.
2. Deployed additional resources such as OpenStack VMs and other platforms for containerized environments will also have an effect on the usage trends.